

Demonstration of Visual Presentation Method for Paranormal Phenomena Through Binocular Rivalry Induced by Dichoptic Color Differences

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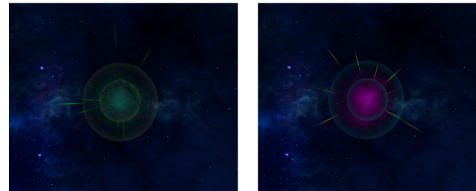
(A) Horror demo



Left eye

Right eye

(B) Super-space demo



Left eye

Right eye

Figure 1: Demonstrations that utilize our mismatched color pair selection methods.

ABSTRACT

Paranormal visual effects, such as spirits and miracles, are frequently depicted in visual games and media design. However, current methods do not express paranormal experiences as aspects of the sixth sense. We propose utilizing binocular rivalry to provide a new visual presentation method by displaying different images in each eye. In this study, we developed a color selection method based on experimentally gathered data to choose a secondary color to display to the other eye when given an initial input color. We then applied this method to create two demonstrations¹. Our approach aims to deliver a more realistic visual experience of paranormal phenomena, thereby enhancing user satisfaction while simultaneously simplifying the design process for designers who utilize our proposed method.

Index Terms: Visual effect, virtual reality, augmented reality, paranormal sensation, binocular rivalry.

1 INTRODUCTION

Paranormal phenomena, such as spirits and magic, exist in dimensions beyond our ordinary perception and have influenced human culture for centuries, continuing to influence our lives [7].

Paranormal phenomena are also frequently featured in video games and other media works. In textual descriptions, spirits are commonly described as “invisible” or possessing “visual elusiveness,” thereby evoking an otherworldly sense of the paranormal [4]. In contrast, current visual techniques for depicting paranormal effects, using transparent materials or shaking visuals [5], offer only symbolic representations.

Given that paranormal experiences resemble a “sixth sense,” existing methods fall short of capturing the cognitive dimension of

such phenomena, and thus fail to deliver an authentic experience. To address this limitation, we explore whether any method can effectively convey the elusive sensation associated with paranormal perception. As a result, we focus on binocular rivalry, which has the potential to offer elusive perceptions both visually and cognitively. As illustrated in Fig. 2, binocular rivalry occurs when dissimilar images are presented to each eye. Instead of merging into a single coherent perception, the two images compete for perceptual dominance: one image can dominate conscious awareness for several seconds, then be supplanted by the previously suppressed rival image [1].

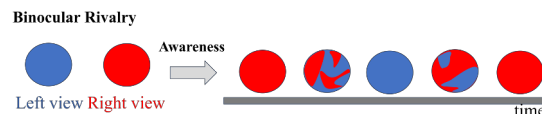


Figure 2: The imagery of binocular rivalry.

A key characteristic of binocular rivalry is its susceptibility to attention: to some extent, one can influence which image dominates, although immediate or sustained control is generally impossible [1]. As a result, the fluctuation between controllable and uncontrollable awareness creates a sense of elusiveness, as if something is visible yet remains indiscernible. Moreover, because binocular rivalry is rarely observed under normal conditions, it offers a unique, other-dimensional visual experience that is particularly well-suited for paranormal visual presentations.

Our previous study demonstrated that displaying mismatched images to each eye can induce a realistic paranormal sensation [2]. Such mismatches may involve shape, color, size, lighting, and other factors. However, the previous work merely introduced the concept without clarifying how these varying visual elements influence paranormal sensations. Moreover, it did not propose a detailed method for designing image discrepancies or applying binocular rivalry. In this research, we explored the effect of color differences between the images presented to each eye on paranormal sensations. Furthermore, we propose a method for selecting mismatched colors and employ it to create two demonstrations (Fig. 1).

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¹Link to a demo video: <https://youtu.be/aNYcILmFIE>

2 METHODS

In our previous experiment [3], we investigated artificially induced color differences between the two eyes to induce binocular rivalry, which could provide more realistic paranormal experiences. This result suggests that our proposed method is well-suited for scenarios involving magic effects, ghosts, and other paranormal elements in video works.

Additionally, in that earlier study, we examined how variations in color difference affect the paranormal sensations and the visual discomfort arising from binocular rivalry. The results revealed that larger color differences not only led to stronger paranormal sensations but also induced visual discomfort. To balance these two factors, we identified a suitable range of color differences that ensures acceptable levels of visual discomfort while maintaining the paranormal sensation. The suitable range varies depending on the sRGB properties of each color. We employed the CIEDE2000 formula [6] to determine this range of color differences ΔE_{00}^* . When red (255,0,0) was selected, the suitable range of color differences ΔE_{00}^* was 22.50 to 30.50. Similarly, for green (0,255,0) and blue (0,0,255), the suitable ranges of ΔE_{00}^* were 22.20 to 32.92 and 18.65 to 0.73, respectively. Building on these results, we developed a method to select the appropriate secondary color for presenting to the other eye when given an input color.

- Designer can select one color for displaying to one eye, then We randomly sampled 10,000 color points within the sRGB color space and then converted these points to the CIELAB color space.
- Our goal is to determine a color for the other eye that balances paranormal sensation and visual discomfort. We recalculate the suitable range based on the RGB proportions of the first color, ensuring that the range aligns with the first color's natural properties. Let (r, g, b) denote the first selected color in sRGB space, and (r_{q1}, g_{q1}, b_{q1}) , (r_{q3}, g_{q3}, b_{q3}) be the experimentally determined suitable range for each color component. The relative proportions of each color component are calculated as follows: $r_{ratio} = \frac{r}{r+g+b}$, $g_{ratio} = \frac{g}{r+g+b}$, $b_{ratio} = \frac{b}{r+g+b}$. In cases where $r+g+b=0$ (i.e., black), we set $r_{ratio} = g_{ratio} = b_{ratio} = \frac{1}{3}$ to avoid division by zero. Next, we calculate the new suitable color difference range ΔE_{q1} to ΔE_{q3} by: $\Delta E_{q1} = (r_{ratio} \cdot r_{q1} + g_{ratio} \cdot g_{q1} + b_{ratio} \cdot b_{q1})$, $\Delta E_{q3} = (r_{ratio} \cdot r_{q3} + g_{ratio} \cdot g_{q3} + b_{ratio} \cdot b_{q3})$.
- In consideration of design diversity, we selected color points from the sampled set whose color differences ΔE_{00}^* lie between ΔE_{q1} and ΔE_{q3} . We then sorted these points by their color difference from the first color and presented the top 100 candidates to the designer.

By the method described above, we implemented a function that automatically selects suitable colors for presentation in the other eye, given a designer-specified input color. Once a color is chosen for one eye, our function automatically generates candidate colors for binocular rivalry. The example of this function is illustrated in Fig. 3. In this example, we selected the RGB value (40,60,120) as the input (Fig. 3 B). Our function then provided a series of candidate colors, visualized on a hue-based color wheel (Fig. 3 A). From among these candidates, we ultimately chose (141,55,78) as the color to be displayed in the other eye (Fig. 3 C).

3 USER EXPERIENCE

Our proposed methods aim to create a more realistic display of paranormal sensations by presenting different images to each eye, thereby inducing binocular rivalry. Moreover, we have designed an automatic method for selecting mismatched color pairs to provide

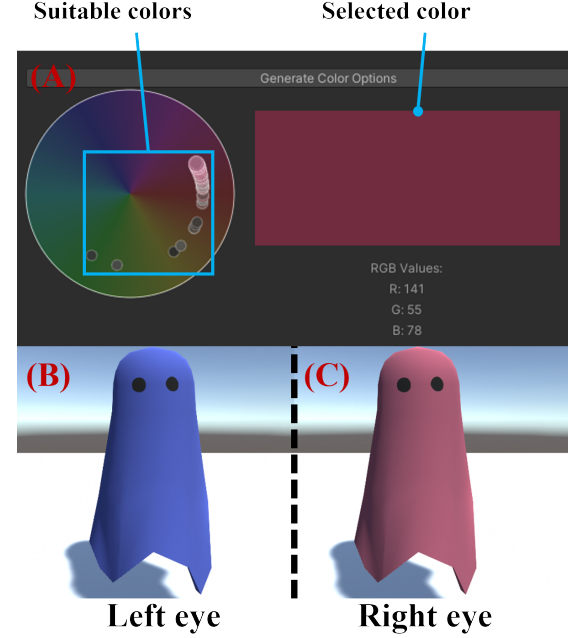


Figure 3: (A) Color Selector UI: The left side displays candidate suitable colors, while the right side shows the currently selected color. (B) A ghost avatar is applied using the first color selected by the user. (C) A ghost avatar is applied using the currently selected color.

color options while balancing visual discomfort. At the IEEE VR research demo, we plan to exhibit two experiences: one demonstrating the color selection method (Fig. 3) and another showcasing the paranormal sensations provided by mismatched colors displayed to both eyes (Fig. 1).

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